PSTAT 5A Quiz 3 Solutions

The Santa Barbara Department of Transportation conducted a study that found **20 percent** of households in Santa Barbara have an e-bike. We select eight households in Santa Barbara at random and are interested in the number that have an e-bike.

a) Define the random variable of interest, X.

Let X be the number of households (among the eight selected) that have an e-bike.

b) State the distribution of X. Show your work by checking the relevant criteria.

This is a binomial setting because:

1. There is a fixed number of trials: $n = 8$ households.
2. Each trial has two outcomes: a household has an e-bike (success) or does not (failure).
3. The probability of success is constant: $p = 0.20$ for each household.
4. The trials are assumed independent, as appropriate for the random sampling model.

Therefore, $\boxed{X \sim \text{Binomial}(8, 0.20)}$.

c) Calculate the probability that exactly 1 household in Santa Barbara has an e-bike.

Using the binomial probability formula,

$$\begin{aligned} P(X = k) &= {}_nC_k\, p^k (1-p)^{n-k}, \\ P(X = 1) &= {}_8C_1 (0.20)^1 (0.80)^{8-1} \\ &= 8(0.20)(0.80)^7 \\ &= \boxed{0.33554432 \approx 0.3355}. \end{aligned}$$

d) Calculate the standard deviation of X.

For a binomial random variable, $\text{Var}(X) = np(1-p)$. Thus,

$$\begin{aligned} \text{Var}(X) &= 8(0.20)(0.80) = 1.28, \\ \text{SD}(X) &= \sqrt{\text{Var}(X)} = \sqrt{1.28} = \boxed{1.1314 \text{ (approximately)}}. \end{aligned}$$